# Övning 1

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Frilägging och snittning i stänger

2.1: 1, 4, 16, 1(DE), 15

## Viktiga begrepp

- Normalkraft (N) [N]
  - Inre kraft mellan två snittade ytor
- Normal spänning ($\sigma$) [Pa] [$N/m^2$]
  - Normalkraften aggerande på en yta

$$\sigma = \frac{N}{A} \quad [Pa] \text{ eller } [N/m^2]$$

- Deformation ($\delta$) [m]
  - Förlängning av tex. en stång
- Förskjutning (u) [m]
  - Hur mycket en punkt förflyttats
- Töjning ($\varepsilon$) [-]
  - Förlängning av en enskild punkt

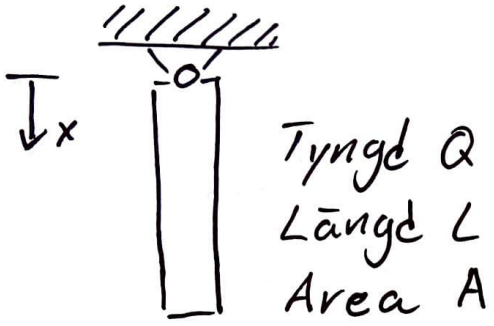
$$\varepsilon = \frac{du}{dx}$$

$$\delta = \int_0^x \varepsilon(x)\,dx$$

- Hookes lag

$$\sigma = E\varepsilon$$

# 2.1.1

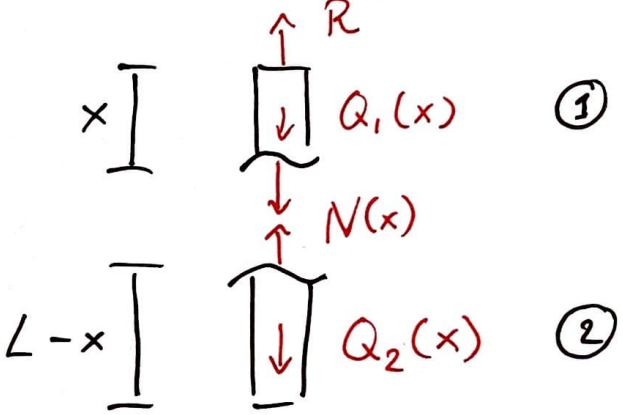



Sökt: Normalspänning i ett tvärsnitt

## Frilägg




## Snitta vid x




## Jämvikt

② ↓: $Q_2(x) - N(x) = 0$

① ↓: $N(x) + Q_1(x) - R = 0 \overset{R=Q}{\Rightarrow} N(x) = Q - Q_1(x)$

## Bestäm $Q_1(x)$

$Q_1(x) = a + bx$ Linjär variation av tyngd

$Q_1(0) = 0 \Rightarrow a = 0$

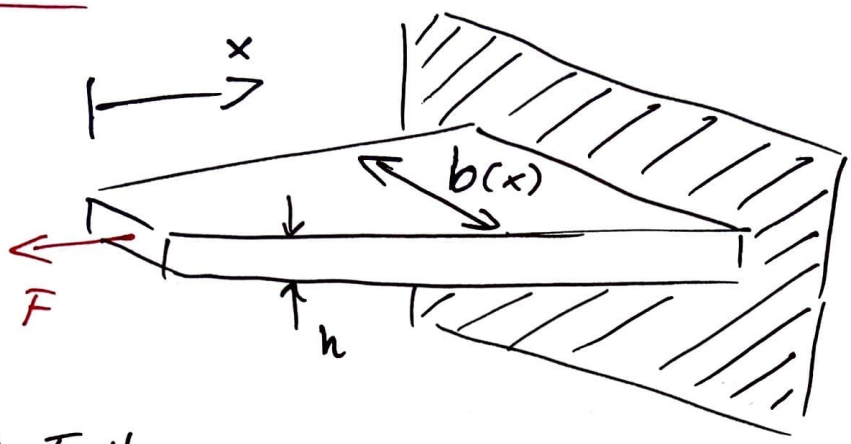
$Q_1(L) = Q \Rightarrow b = \frac{Q}{L} \Rightarrow Q_1(x) = Q\frac{x}{L}$

## Sätt in $Q_1(x)$ i normalspänningen

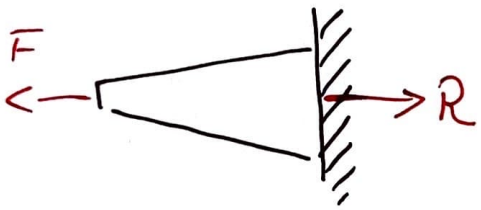
$N(x) = Q\left(1 - \frac{x}{L}\right) \Rightarrow \boxed{\sigma(x) = \frac{Q}{A}\left(1 - \frac{x}{L}\right)}$

# 2.1.4

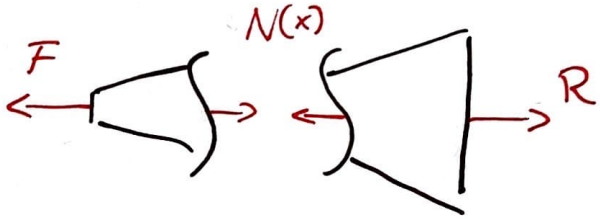
**Sökt:**
Normal spänning
$\sigma(x)$




① Frilägg




② Snitta




③ Jämvikt

$\rightarrow$: $N - F = 0 \qquad N = F$

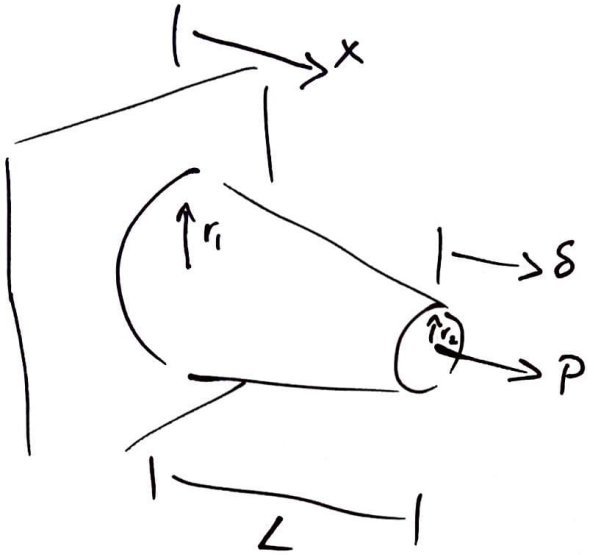
④ Ställ upp uttryck för normal spänning

$$\sigma(x) = \frac{N(x)}{A(x)} = \frac{F}{A(x)}$$

$$A(x) = h\,b(x) = h\,b_0\left(1 + \frac{x}{L}\right)$$

$$\boxed{\sigma(x) = \frac{F}{h\,b_0}\,\frac{1}{\left(1 + \frac{x}{L}\right)}}$$

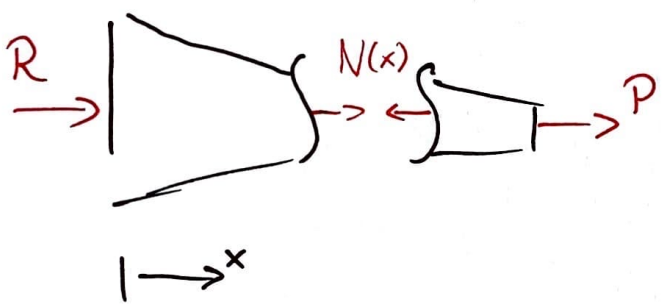
2.1.16




## Elasticitetsmodul E

**Sökt: Kraft - deformations sambandet $\delta(P)$**

① Frilägg & snitta




② Jämvikt

$\rightarrow$: $N(x) - P = 0$

$N(x) = P$

Konstant

③ Spänning

$$\sigma(x) = \frac{N(x)}{A(x)} \qquad A(x) = r(x)^2 \pi$$

$r(x) = a + bx$ Linjärt samband

$$r(x) = r_1 + \frac{r_2 - r_1}{L} x$$

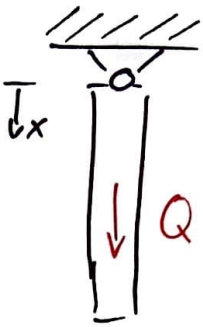
④ Töjning

$$\varepsilon(x) = \frac{\sigma(x)}{E} = \frac{P}{E\pi\left(r_1 + \frac{r_2 - r_1}{L} x\right)^2}$$

⑤ Deformation

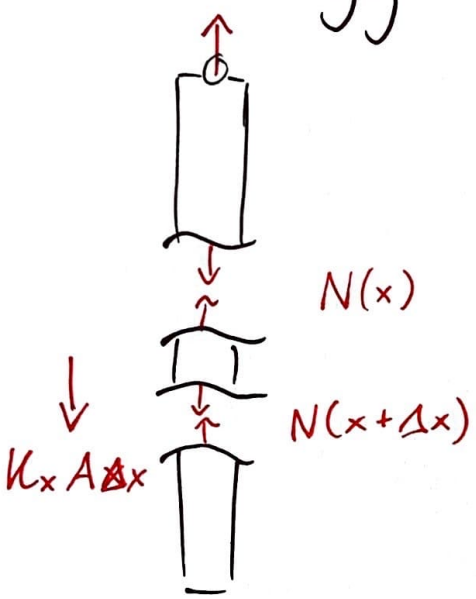
$$\delta = \int_0^L \frac{P}{E\pi\left(r_1 + \frac{r_2 - r_1}{L} x\right)^2} = \boxed{\frac{PL}{E\pi r_1 r_2}}$$

# 2.1.1 with DE




Sökt:
Normalspänningen
$\sigma(x)$

① Frilägg




② Jämvikt

$\downarrow$: $N(x+\Delta x) - N(x) + K_x A \Delta x = 0$

$$\underbrace{\frac{N(x+\Delta x)-N(x)}{\Delta x}}_{\frac{dN}{dx}} + K_x A = 0$$

$$\frac{dN}{dx} + K_x A = 0$$

$$\left(K_x = \rho g = \frac{Q}{AL}\right)$$

③ Lös Jämvikts ekvationen

$$\frac{dN(x)}{dx} + \frac{QA}{AL} = 0 \Rightarrow \frac{dN(x)}{dx} = -\frac{Q}{L}$$

$$N(x) = -\frac{Q}{L}x + C$$

④ Randvillkor

$N(L) = 0 \quad N(0) = Q \Rightarrow C = Q$

$$N(x) = Q\left(1 - \frac{x}{L}\right)$$

$$\sigma(x) = \frac{Q}{A}\left(1 - \frac{x}{L}\right)$$

# 2.1.15

Samma stång som 2.1.4

$$\sigma(x) = \frac{Q}{A}\left(1 - \frac{x}{L}\right)$$

① Töjning (Hookes lag)

$$\varepsilon = \frac{\sigma}{E} \qquad \varepsilon(x) = \frac{Q}{EA}\left(1 - \frac{x}{L}\right)$$

② $$\delta = \int_0^L \varepsilon(x)\,dx = \left[\frac{Q}{EA}\left(x - \frac{x^2}{2L}\right) + C\right]_0^L$$

$$= \frac{Q}{EA}\left(L - \frac{L^2}{2L}\right) = \frac{QL}{2EA}$$

$$\boxed{\delta = \frac{QL}{2EA}}$$